

Branch-wise versus clade-grouped gene-family evolution on the archaeal tree

Reconciliation of 7,059 gene families on the 257-taxon archaeal species tree, Euryarchaeota root

Reconciling each gene family against the species tree under a duplication–transfer–loss (DTL) model requires a choice of how finely rates may vary across the tree. We compared two parameterisations: the *clade-grouped* model, in which all branches of a major clade share a single set of duplication, loss and transfer rates (and origination weight), and a *fully branch-wise* model, in which every branch carries its own rates. As a control, our GPU reconciliation engine reproduces the reference per-family likelihoods to within Pearson $r = 0.999998$, so the comparison reflects the rate model, not the implementation.

Relaxing the clade constraint improves the fit by $\sim 10,900$ log-likelihood units (Table 1) while leaving the inferred root unchanged: Euryarchaeota remains the maximum-likelihood rooting under both models.

A model with thousands of free rates risks describing noise rather than signal, so we tested it by cross-validation (Figure 1). We estimated the rates from a random half of the gene families and used them to predict the likelihood of the other half, which took no part in fitting. The branch-wise model predicts these withheld families better than the clade-grouped model, by +1.66 log-likelihood units per family — as much as on the families used for fitting (+1.49 per family). Had the additional parameters been absorbing noise they would have *degraded* prediction on the withheld families; instead they improve it by the same margin in and out of sample. The among-branch rate variation the model adds is therefore real, not an artefact of over-parameterisation.

That variation has a clear and interpretable structure (Figure 2). Duplication and transfer rates are essentially *clade-constant*: within each named clade the per-branch estimates scarcely depart from a single value (clade membership explains $R^2 \approx 0.94$ of their variance). Loss is markedly less clade-structured ($R^2 \approx 0.38$): its extra variation is distributed across branches rather than captured by clade identity. Biologically, gene gain (duplication, transfer) is well summarised by one rate per clade, whereas gene loss is more branch-idiosyncratic.

The reconstructed ancestral genome depends far more on the origination treatment than on the rate granularity (Figure 3). Of the 7,059 families, a model with *uniform* origination places only 96 at the last archaeal common ancestor — a flat origination prior makes the root no likelier a birthplace than any other branch. Allowing origination to be *fitted* raises this almost eight-fold — to 772 under clade-grouped rates and 903 under the branch-wise model — as gene births concentrate near the root. The further step from clade-grouped to branch-wise rates adds comparatively little (772 to 903; the two share 722 families, Jaccard 0.76, with per-family root-origination probabilities correlated $r = 0.96$). So the size of the inferred ancestral genome is set chiefly by the origination model, and the branch-wise rates refine rather than reshape it.

The same reconciliation reconstructs the gene complement of *every* ancestral node, not only the root: summing each family’s posterior probability of being present at a node gives that node’s expected genome size. Figure 4 shows these sizes at the ancestor of each named clade under the branch-wise model. The last archaeal common ancestor (root) carries $\approx 1,350$ families in expectation (the stricter $PP \geq 0.5$ families-at-root count of Figure 3 is 903); several deep ancestors — DPANN, Asgard, TAC — are reconstructed larger still, and the shallowest sub-lineages (Pacearchaeota, Nanoarchaeota) smallest. Per-family expected copy numbers at every node are provided as a supplementary table (`per_family_copies_bigtrees_Eury.tsv.gz`, 7,059 families $\times$ 513 nodes).

candidate root	data $\log L$ ($\times 10^3$ nats)			AU support		
	uni. clade	n-u. clade	branchwise	uni. clade	n-u. clade	branchwise
Euryarchaeota	-1645.0	-1642.4	-1631.5	0.65	1.00	1.00
HalobacThermopl	-1644.7	-1647.8	-1635.1	<.05	<.05	0.00
TackA	-1645.1	-1643.5	-1635.5	<.05	<.05	0.00
MHH	-1644.7	-1649.5	-1637.7	0.48	<.05	0.00
Kor	-1645.7	-1648.0	-1637.9	<.05	<.05	0.00
TACA	-1646.1	-1648.4	-1638.4	<.05	<.05	0.00
DPANN	-1645.2	-1644.3	-1639.7	<.05	<.05	0.00
TAC	-1647.0	-1648.9	-1640.0	<.05	<.05	0.00
Asgard	-1647.0	-1649.0	-1641.1	<.05	<.05	0.00
MicraDia	-1645.3	-1646.3	-1642.3	<.05	<.05	0.00
AMD	-1645.7	-1647.6	-1642.5	<.05	<.05	0.00
UndineClu2	-1645.5	-1648.0	-1643.2	<.05	<.05	0.00
Alti	-1645.0	-1648.3	-1643.3	<.05	<.05	0.00
Cluster2	-1645.4	-1648.4	-1643.7	<.05	<.05	0.00
Micra5	-1645.5	-1648.9	-1644.1	<.05	<.05	0.00

Table 1: All 15 candidate roots under the three models: *uni. clade* = clade rates with *uniform* origination ($\approx$ AleRax’s no-origination BIC-best model); *n-u. clade* = clade rates with *fitted* origination; *branchwise* = per-branch rates with fitted origination. Log-likelihoods are gpurec marginal values (directly comparable); AU are scale-invariant approximately-unbiased multiscale-bootstrap supports (10^6 replicates per scale). The uniform-origination model is a genuine near-tie: Euryarchaeota and MHH are each selected in a large fraction of bootstrap replicates (bootstrap proportion 0.57 and 0.41; AU 0.65 and 0.48), a clean {Euryarchaeota, MHH} confidence region (no other root reaches AU 0.05). Adding fitted origination (same clade rates) makes Euryarchaeota decisive: it is selected in *every* replicate at every scale (it won all 10^7 resamples; bootstrap proportion 1.0, hence AU = 1.0) under both the clade-grouped and branch-wise models — the rooting comes from the *origination* term, not the rate granularity. (At a saturated bootstrap proportion of 1.0 the multiscale-AU regression is degenerate and returns a spurious ~ 0.63 ; we report the convention AU = 1.0 for a tree selected in all replicates.) The branch-wise rates then improve the fit at every root (by +10,888 nats at Euryarchaeota over the non-uniform clade model) without changing the rooting; on families withheld from fitting they predict +1.66 per family better (Figure 1).

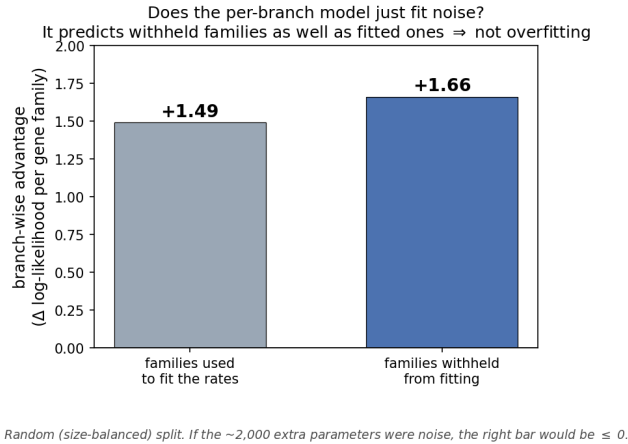


Figure 1: Cross-validation against over-parameterisation. The branch-wise model’s predictive advantage over the clade-grouped model, per gene family, on families used to fit the rates versus families withheld from fitting. Both are positive — the per-branch model predicts the withheld families better — so the added rates are real signal. Had they been fitting noise, the right-hand bar would be ≤ 0 .

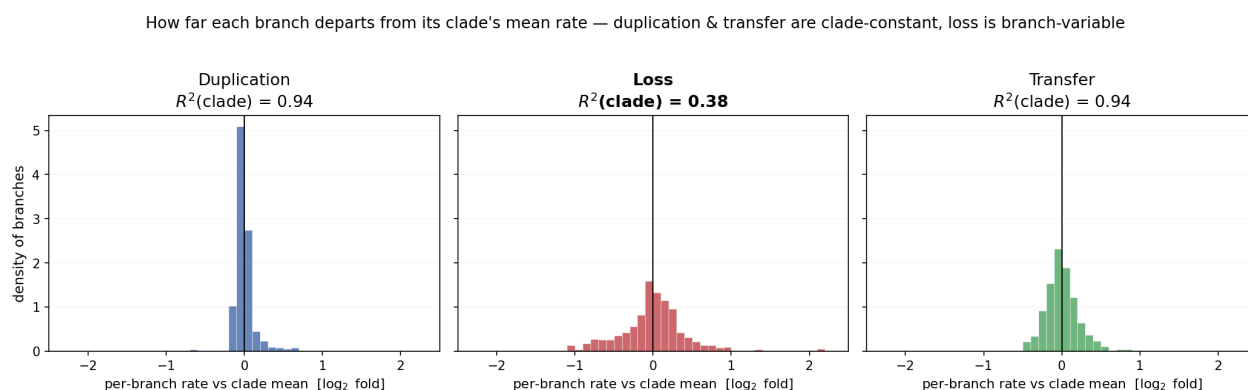


Figure 2: Departure of each branch's fitted rate from its clade's mean rate ($\log_2$ fold; shared axes). Duplication and transfer form a sharp spike at zero — the per-branch estimates scarcely leave their clade mean, so clade membership explains $R^2 \approx 0.94$ of the variance — whereas loss is broad ($R^2 \approx 0.38$), i.e. branch-specific. Gene gain (duplication, transfer) is well summarised by one rate per clade; gene loss is not.

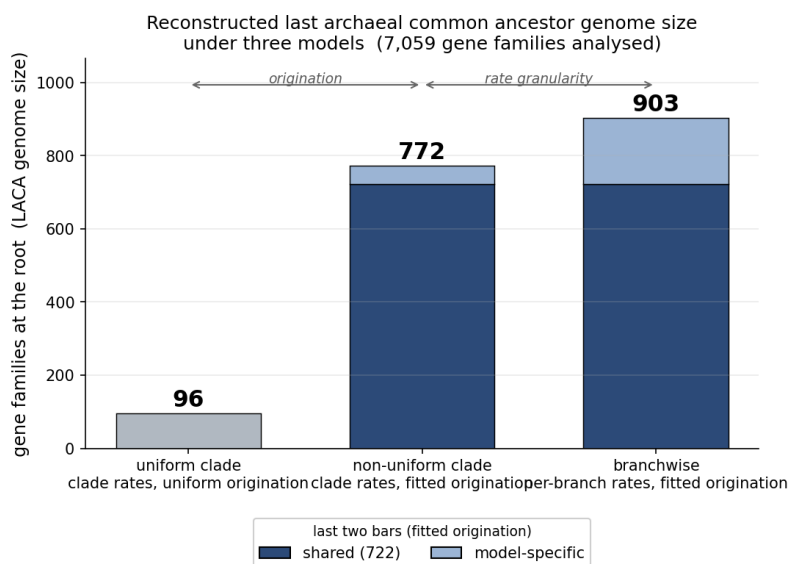


Figure 3: Reconstructed LACA genome size under three models, isolating the two effects. Uniform origination places only 96 families at the root; fitting origination with the same clade-grouped rates raises it to 772 (the origination effect), and the branch-wise rates add a further 131 to 903 (the rate-granularity effect). The two fitted-origination models share 722 of these families (dark), differing only in small model-specific remainders (light). Origination dominates; rate granularity refines. (Of 7,059 families.)

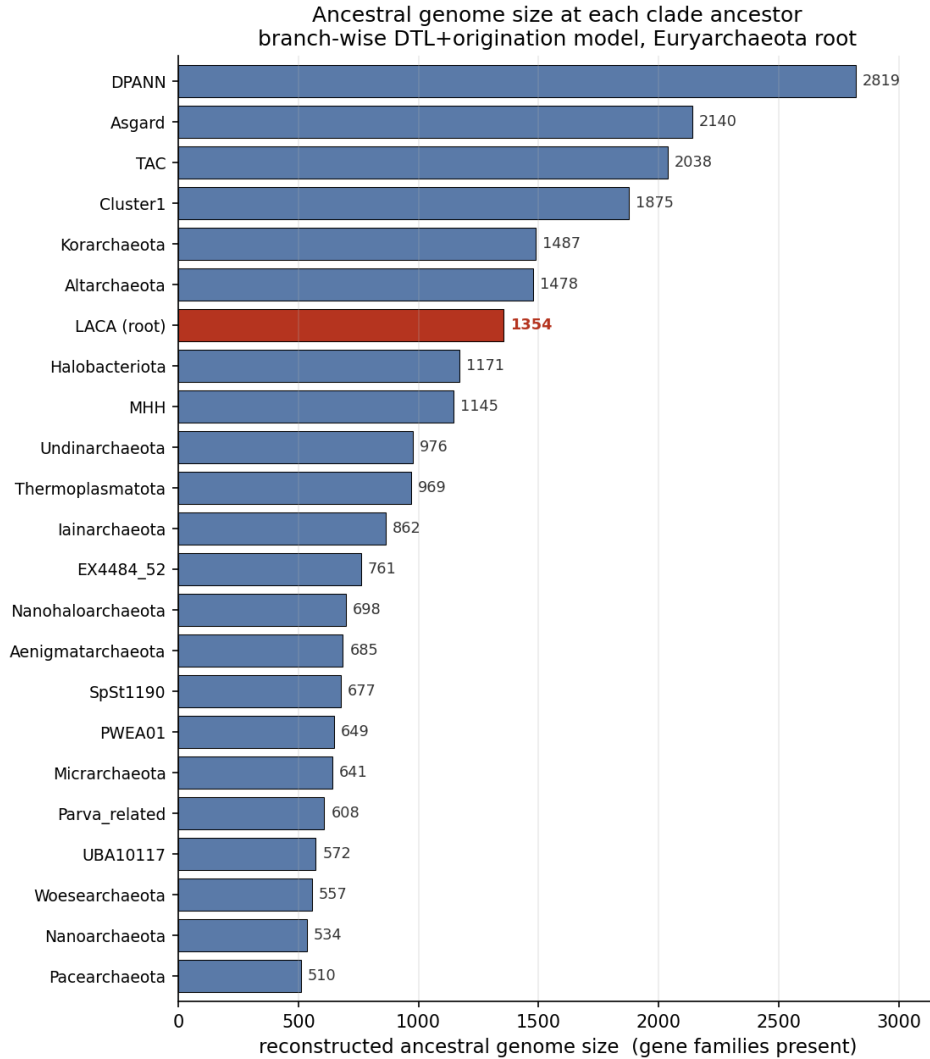


Figure 4: Reconstructed ancestral genome size — the expected number of gene families present (summed posterior presence) — at the ancestor of each named clade, under the branch-wise DTL+origination model on the Euryarchaeota-rooted tree. The root ancestor (LACA, highlighted) carries $\approx 1,350$ families in expectation; the thresholded ($PP \geq 0.5$) LACA count is the 903 of Figure 3. Reconstructed by stochastic backtracking of the reconciliation (100 samples per family).